

An Island Reshaped from Within: 34 Years of Coastal and Inland Water Change on Belitung Island, Indonesia, Measured in R

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Abstract

Belitung Island in the Java Sea has been mined for tin since the 1800s, and residents report that the sea is eroding the land. Eight hand-drawn land outlines were measured, 1990 to 2024, in R with the package `simple features (sf)`, in Universal Transverse Mercator (UTM) zone 48 South, and fitted trends with a linear model (`lm`). Land area falls 0.80 square kilometres (km²) per year, but R-squared (R^2) is 0.18 and the p-value 0.29, so I cannot say the island is shrinking. Water bodies inside it rise from 1,294 to 2,972, about 57 a year ($R^2 = 0.62$, $p = 0.020$). Net loss is 38.1 km², 0.81 percent, and 1,273 hectares (ha) of it lies over one kilometre (km) inland, beyond the reach of direct marine inundation, which cannot explain water that far inland; abandoned open tin pits filling with rainwater can. On Belitung the increase in water is concentrated in the island's interior, not at its coastline.

Introduction and Research Questions

What my measurements can answer, not how to reverse erosion.

- ◆ Belitung lies in the Java Sea, between Sumatra and Borneo (Fig. 1).
- ◆ A tin island: an abandoned open pit fills with rain and becomes a kolong lake.
- ◆ Q1. How much land did Belitung lose between 1990 and 2024?
- ◆ Q2. Is the loss at the coast, where the sea reaches, or inland, where it cannot?

Methodology

R and RStudio, package `sf`. All measurement in UTM zone 48 South, European Petroleum Survey Group (EPSG) code 32748.

- ◆ Eight yearly land outlines, digitised from binary water rasters.
- ◆ ArcGIS 10.5: `RasterToPolygon (SIMPLIFY)`, then smoothing by polynomial approximation with exponential kernel (PAEK) at 60 metres (m), in World Geodetic System 1984 (WGS 84).
- ◆ `st_transform()` to metres; ring 1 is the coast, every other ring is inland water.
- ◆ Table 1 lists `st_area()`, `st_length()` and ring counts per year; `lm()` gives slope, R^2 and p .
- ◆ `st_difference()` maps 1990 against 2024; `st_buffer()` splits coast from inland at 1 km.

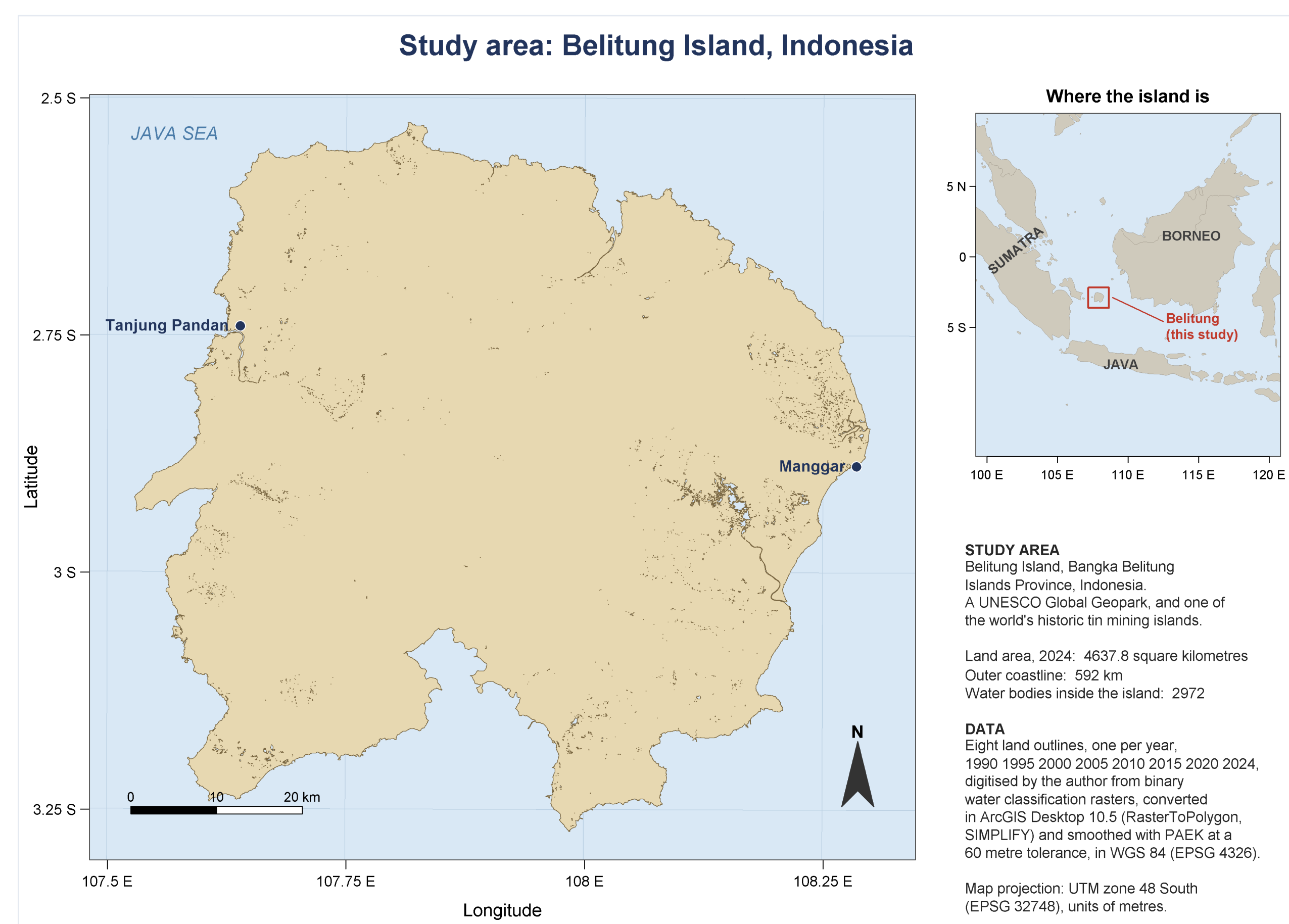


Fig. 1. Map of Belitung Island, Indonesia, drawn from the 2024 land outline, with an inset locating the island in the Java Sea. Land area 4,637.8 km², outer coastline 592 km.

SELECTED REFERENCES

Pekel J.-F., Cottam A., Gorelick N. and Belward A.S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature*, 540, 418-422.
JRC Global Surface Water, transitions layer v1.4 (2021), tile transitions_100E_0Nv1_4_2021.tif, 30 m pixels, 1984 to 2021. European Commission Joint Research Centre.
Pebesma E. (2018). Simple features for R. *The R Journal*, 10(1), 439-446.
Marsellos A.E. (2026). HUSSRP teaching material: Tutorial_CoastalChange_Belitung_OwnShapefiles and Tutorial_CoastalErosion_ShorelineChange. Hofstra University.

Results

Belitung Island only, from the eight outlines digitised here. No Muara Gembong number here; that separate study area is shown in Fig. 4.

- ◆ Land area falls 0.80 km² per year, but $R^2 = 0.18$ and $p = 0.29$, so the decrease is not statistically significant (Fig. 2, left).
- ◆ Water bodies inside the island: 1,294 to 2,972, about 57 a year ($R^2 = 0.62$, $p = 0.020$).
- ◆ 1990 against 2024: 51.0 km² of land became water, 12.9 km² became land (Fig. 3).
- ◆ Net loss 38.1 km², 0.81 percent of the 1990 land area.
- ◆ Of that loss, 3,823 ha lie within 1 km of the 1990 coast, 1,273 ha deeper inland.
- ◆ Fastest of the eight 45-degree sectors measured from the island centroid: the east coast, 3.29 m per year of coastline retreat against an island average of 1.45.

Table 1. The eight digitised Belitung outlines, year by year: land area, coast length, and the number and area of inland water bodies, one water body being one lake or pond mapped inside the coastline. Land area varies by 1.3 percent; water bodies more than double.

Year	Land area (km ²)	Coast (km)	Water bodies	Inland water (km ²)
1990	4,675.9	610	1,294	10.6
1995	4,700.5	586	1,249	8.6
2000	4,667.3	578	1,021	6.4
2005	4,680.1	579	1,058	6.9
2010	4,659.0	581	1,097	7.6
2015	4,650.6	563	1,844	10.9
2020	4,700.5	682	3,133	17.7
2024	4,637.8	592	2,972	17.9

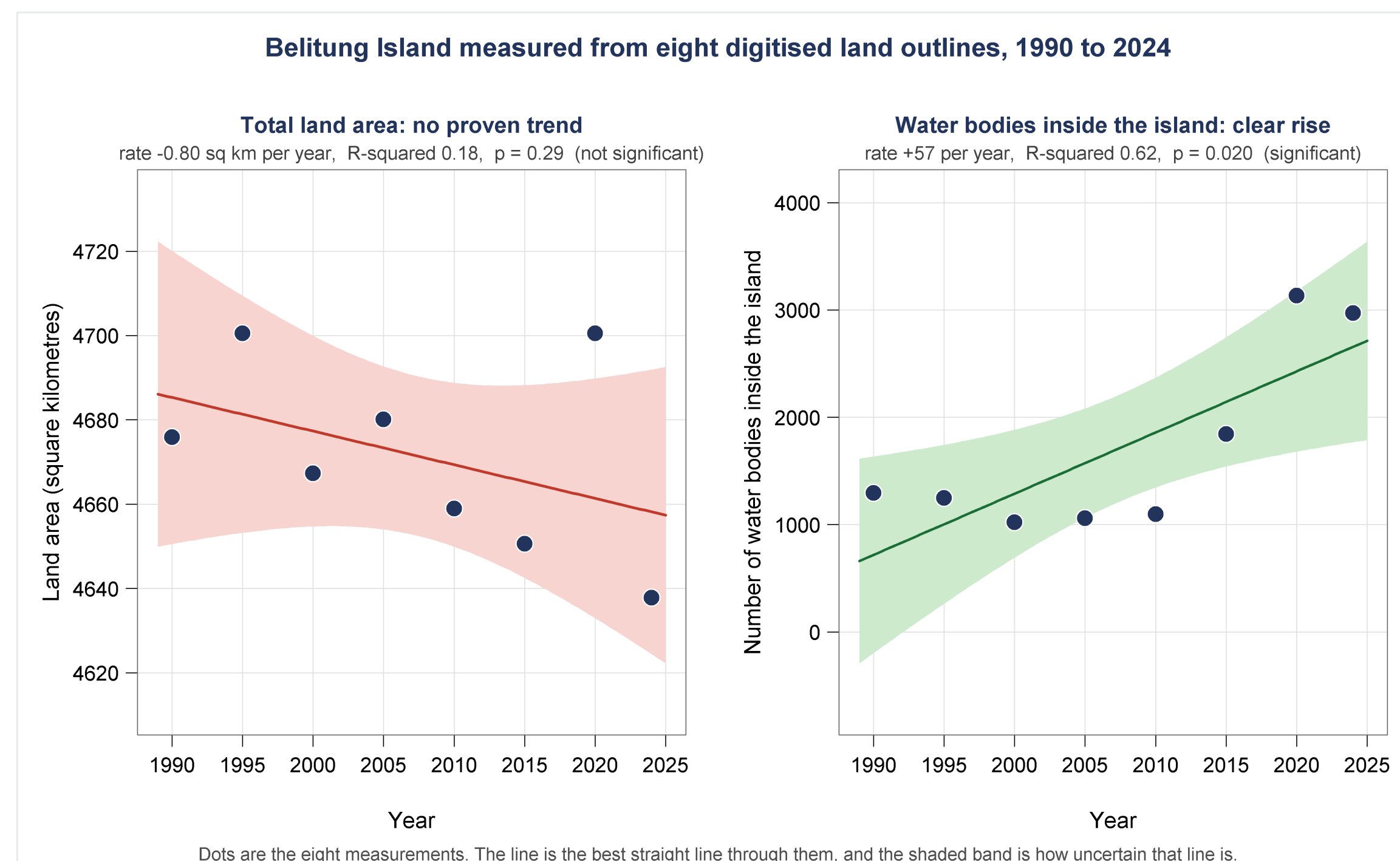


Fig. 2. Scatter plots of the eight years, 1990 to 2024: total land area on the left, inland water bodies on the right, each with a fitted line. The land trend is not statistically significant; the rise in water bodies is.

Discussion and Conclusions

- ◆ The pair is the finding: a land trend that is not statistically significant beside a rising inland water count.
- ◆ 1,273 ha of the new water sits over 1 km inland, beyond the reach of direct marine inundation.
- ◆ Abandoned tin pits fill with rain and become kolong lakes, adding separate water bodies.
- ◆ Conclusion: after a net loss of 38.1 km², the increase in water is concentrated in the island's interior, not at its coastline.

ACKNOWLEDGEMENTS

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Land lost and gained on Belitung Island, 1990 to 2024

Net loss 38.1 square kilometres, which is 0.81 percent of the 1990 land area

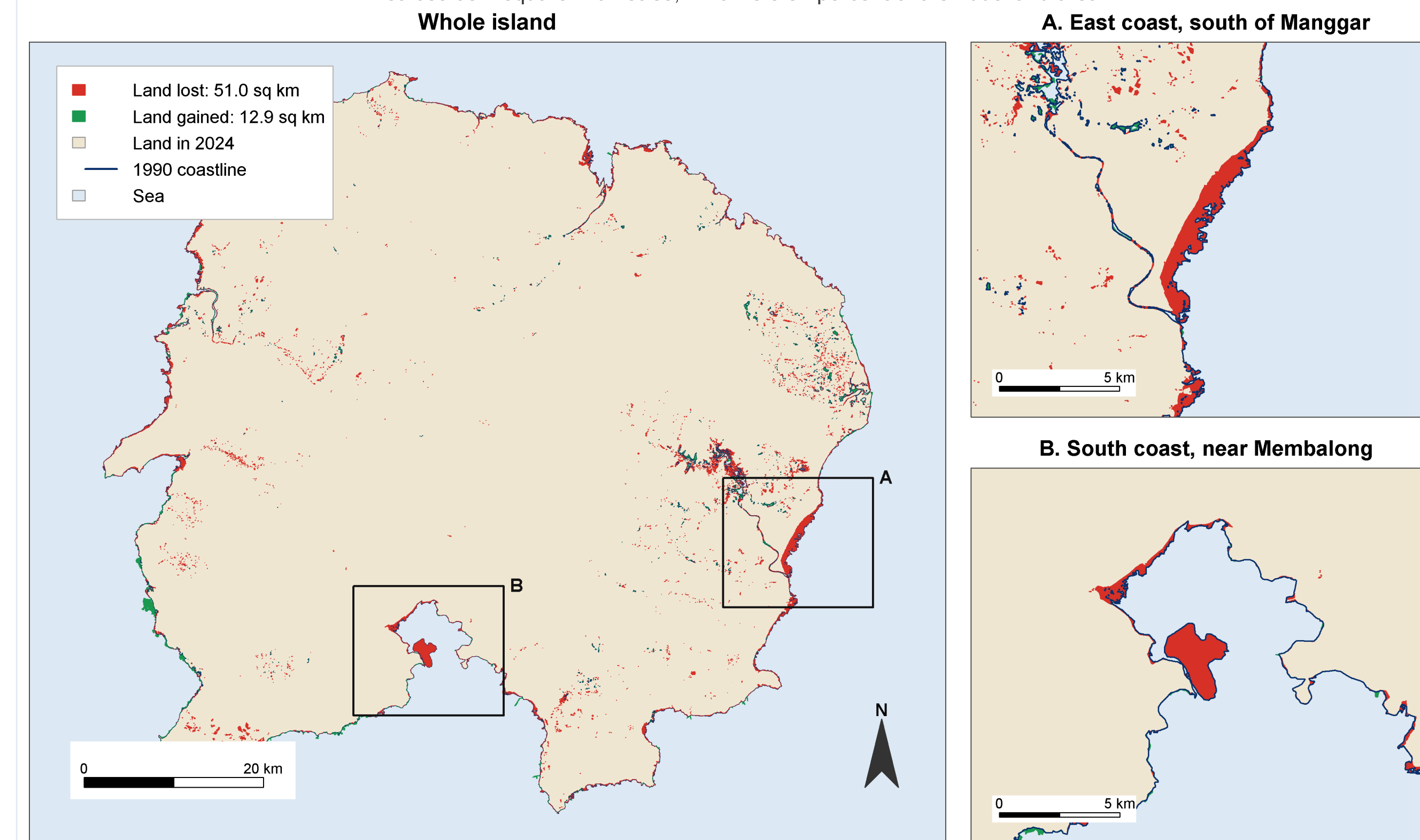


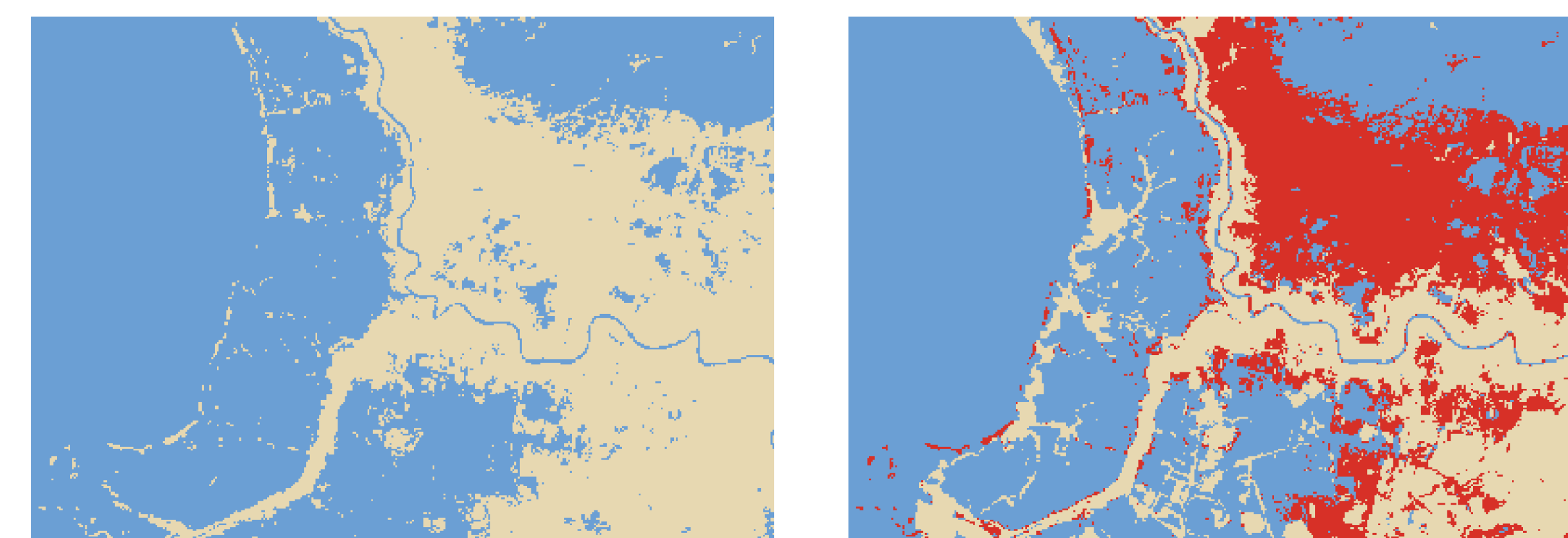
Fig. 3. Map overlaying the 1990 and 2024 outlines of Belitung: red is land that became water, green is water that became land. Red areas are concentrated on the east coast and well inland.

Muara Gembong, Jakarta Bay: the sea has advanced over the land

A separate study area, not Belitung Island

EARLIER (about 1990)

RECENT (about 2020)



Built from JRC Global Surface Water. Red = land that turned into sea.

Fig. 4. Before and after maps of Muara Gembong, Jakarta Bay, a separate study area, not Belitung, from the European Commission Joint Research Centre (JRC) Global Surface Water, 1984 to 2021; red is land that became sea; net loss 1,478 ha (1,842 lost, 364 gained).

LIMITATIONS

- ◆ Outlines are hand-drawn: operator error sits in every Belitung number.
- ◆ PAEK smoothing at 60 m removes any coastal bend under about 60 metres.
- ◆ 1995 and 2020 both give 4,700.5 km²; 2020's coast is 682 km against 563 to 610 km.
- ◆ The Muara Gembong product (Fig. 4) covers 1984 to 2021 in one layer, so no yearly rate.
- ◆ Source satellite images and dates are undocumented, so none is named.
- ◆ No mining permit or pit map was available, so the link to abandoned tin pits is a pattern in the data, not a demonstrated cause.

SIGNIFICANCE

Local accounts on Belitung attribute the island's land loss to the sea. The island's area has barely moved in 34 years, but the water bodies inside it have more than doubled. That points away from sea walls, towards the pits already in the ground.

KEYWORDS AND CONTACT

Belitung Island, Indonesia, coastal change, shoreline digitising, inland water bodies, kolong lakes, tin mining, null result, Muara Gembong, R, `sf`
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